

ONLINE EXAMINATION SYSTEM

1. Introduction

The Online Examination System is a web-based application developed using Python Flask, HTML, CSS, and SQLite. The system enables students to register, log in securely, take online examinations, and receive instant results. The application automates the examination process and eliminates manual evaluation. The project provides an easy-to-use interface for students while maintaining exam records in a database.

2. Technologies Used

Frontend :

- HTML5
- CSS3

Backend:

- Python
- Flask Framework

Database:

- SQLite

Development Tool:

- Visual Studio Code

3. System Requirements

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: 4 GB or above
- Storage: 500 MB free space

Software Requirements

- Python 3.x
- Flask
- SQLite
- Visual Studio Code

4. Module Description

4.1 Student Registration Module

This module allows new users to create an account using a unique username and password.

Features:

- User Registration
- Duplicate Username Validation
- Database Storage

4.2 Login Module

This module authenticates students using their credentials.

Features:

- Username Verification
- Password Verification
- Session Management

4.3 Examination Module

This module displays multiple-choice questions retrieved from the SQLite database.

Features:

- Dynamic Question Display
- Multiple Choice Options
- Answer Submission

4.4 Evaluation Module

This module compares student answers with the correct answers stored in the database.

Features:

- Automatic Evaluation
- Score Calculation
- Accuracy Measurement

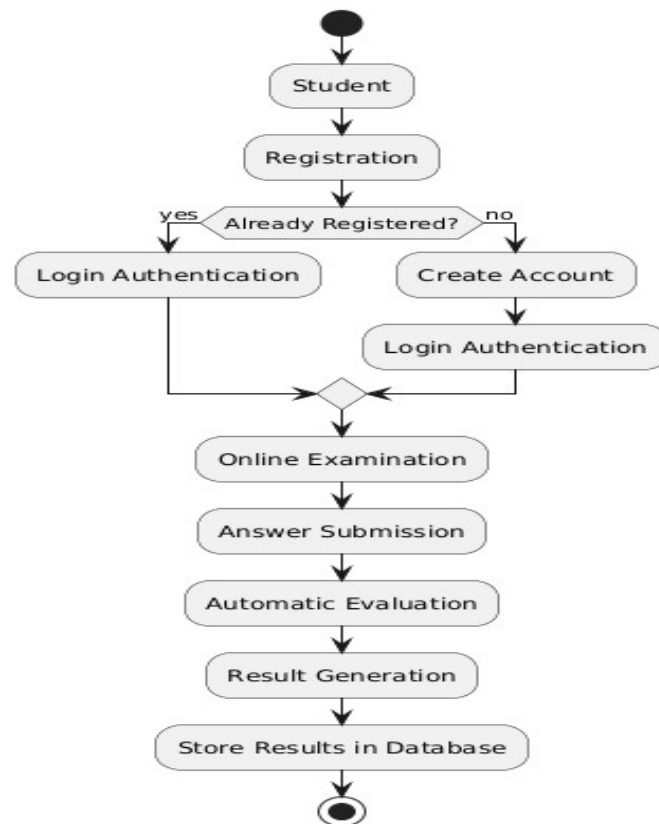
4.5 Result Module

This module displays the examination result immediately after submission.

Features:

- Score Display
- Pass/Fail Status
- Result Storage

5. System Architecture



6. Database Design

Students Table

Field	Type
id	Integer
username	Text
password	Text

Questions Table

Field	Type
id	Integer
question	Text
options	Text
answer	Text

Results Table

Field	Type
id	Integer
username	Text
score	Integer

7. Workflow

1. Student opens application.
2. Student registers an account.
3. Student logs in using credentials.
4. Questions are fetched from database.
5. Student selects answers.
6. Exam is submitted.
7. System evaluates responses.
8. Result is displayed.
9. Score is stored in database.

8. Future Enhancements

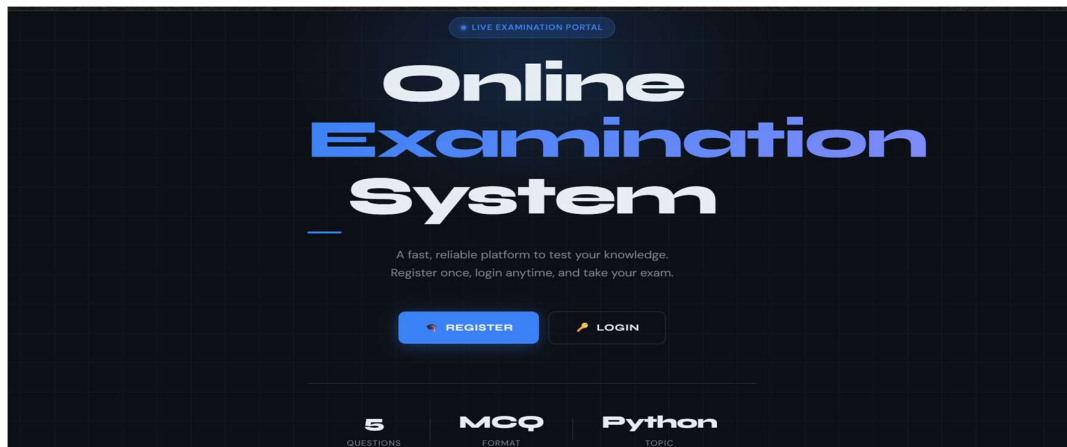
- Admin Panel
- Question Management System
- Timer-based examination
- Result Analytics Dashboard
- Email Notifications

9. Conclusion

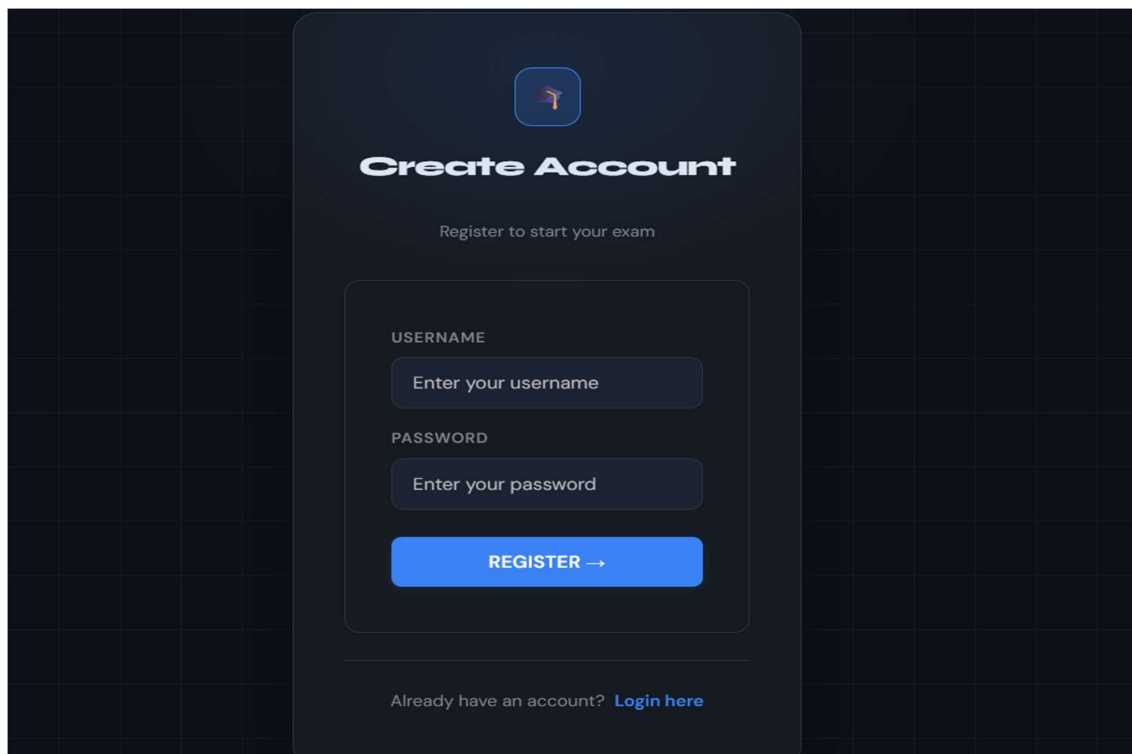
The Online Examination System successfully provides a platform for conducting examinations online. The project automates registration, authentication, examination, evaluation, and result generation processes. The system improves efficiency, reduces manual work, and provides a user-friendly examination environment for students.

10. Output

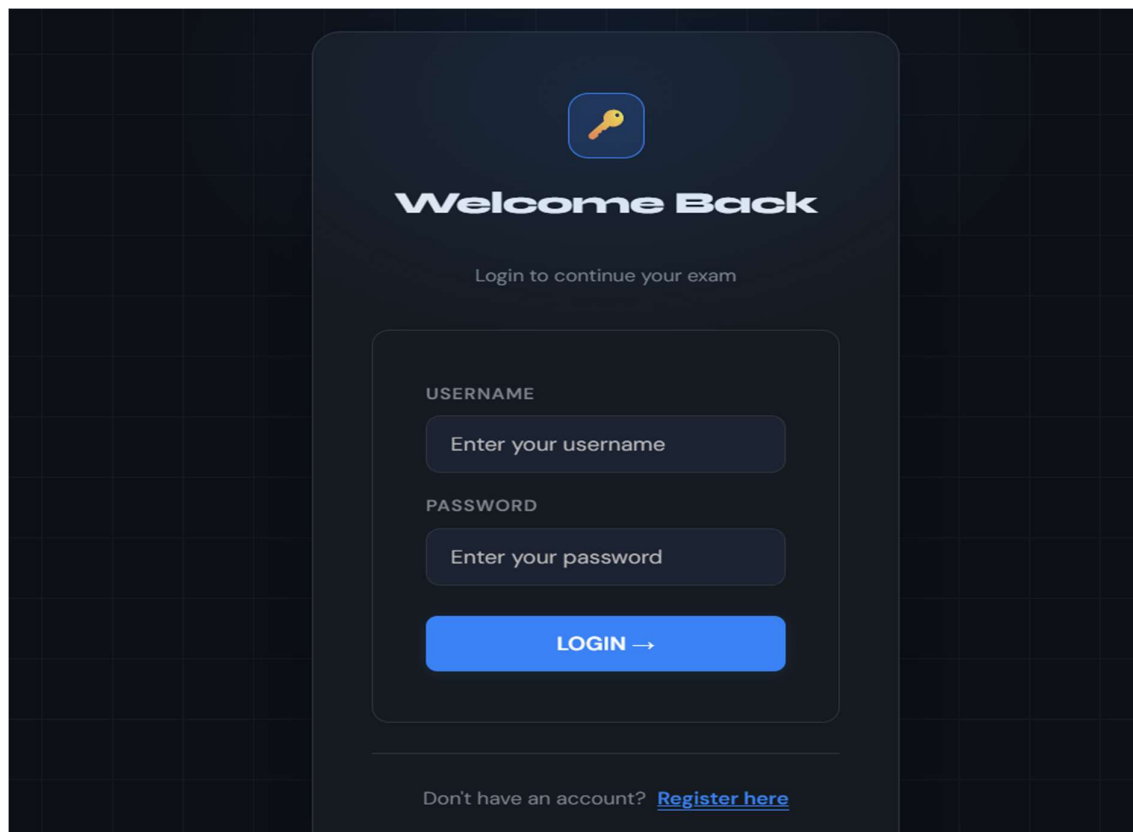
Home Page:



Register page:

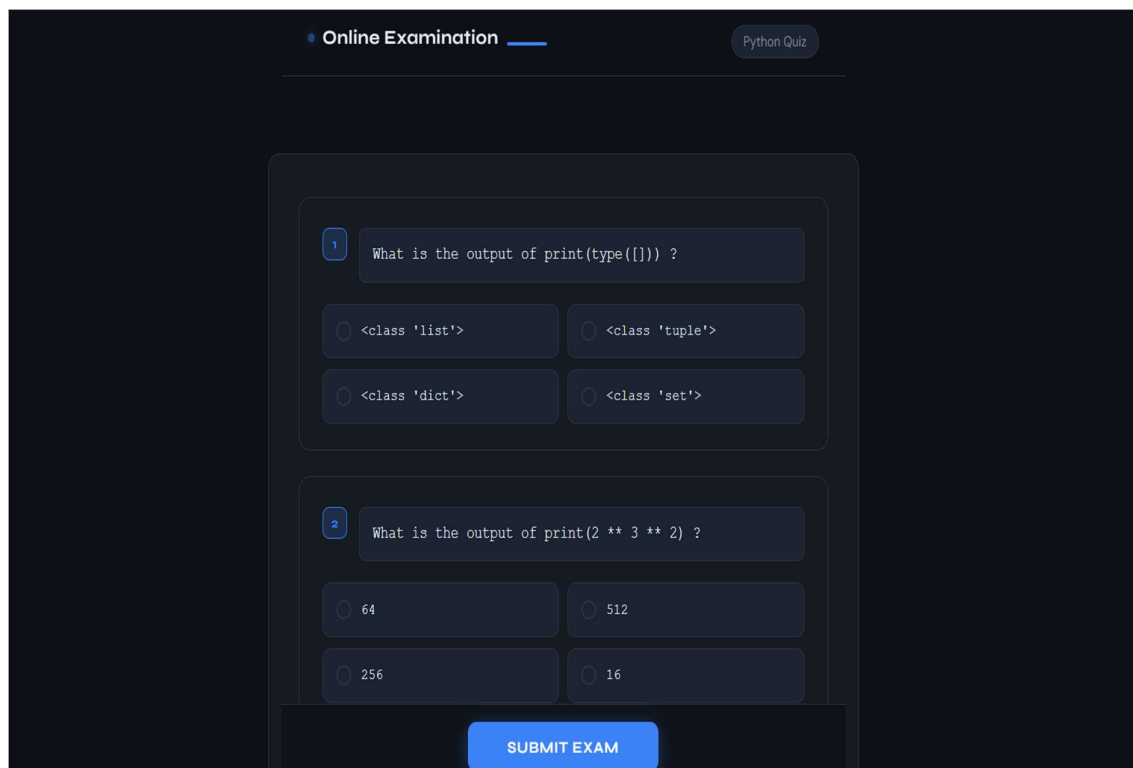


Login page:



The login page features a dark blue background with a subtle grid pattern. A central white card contains the login form. At the top of the card is a yellow key icon inside a blue rounded square. Below the icon, the text "Welcome Back" is displayed in a large, bold, white font. Underneath, the text "Login to continue your exam" is shown in a smaller white font. The form includes two input fields: "USERNAME" with the placeholder "Enter your username" and "PASSWORD" with the placeholder "Enter your password". A prominent blue button with the text "LOGIN →" is positioned below the password field. At the bottom of the card, a link "Don't have an account? [Register here](#)" is provided in white text.

Exam page:



The exam page has a dark blue background. At the top, there is a header with "Online Examination" and a "Python Quiz" tab. The main content area contains two questions. Question 1 asks "What is the output of print(type([])) ?" and offers four radio button options: "<class 'list'>", "<class 'tuple'>", "<class 'dict'>", and "<class 'set'>". Question 2 asks "What is the output of print(2 ** 3 ** 2) ?" and offers four radio button options: "64", "512", "256", and "16". A blue button labeled "SUBMIT EXAM" is located at the bottom center of the page.

Result page:

